

What Powers You

March 14, 2025

National Coordinating Office
AI Action Plan Attn: Faisal D'Souza
2415 Eisenhower Avenue
Alexandria, VA 22314

Dear Mr. Faisal:

Bloom Energy respectfully submits the following comments for the Request for Information released February 6, 2025, in support of Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence).

Bloom Energy fully supports policies that unleash American innovation in the development of AI. The United States must have a robust, resilient and modern electric delivery system to build the infrastructure needed to prevail in the AI Race against China.

Success will require an “All of the Above” approach to energy delivery, incorporating innovative solutions such as onsite distributed energy resources (DER) like stationary fuel cells and microgrids. Any final AI Action Plan must address urgent and timely power needs, including traditional and innovative utility approaches alongside DERs and microgrids.

Bloom Energy is a U.S.-based manufacturer of solid oxide fuel cell (“SOFC”) technology. Our SOFCs utilize an electro-chemical process that uses natural gas and other feedstocks to generate electricity without combustion. Stationary onsite fuel cells can reduce load on the electricity system and provide much needed grid capacity. In addition to avoiding line losses and other risks created by electric transmission, these distributed energy resources can simultaneously power critical facilities, like traditional data centers and larger hyperscale data centers, assets critical to winning the AI Race.

Bloom Energy is committed to manufacturing in the United States and has invested heavily in capacity expansion. In August 2022, the company unveiled a state-of-the-art 164,000 square foot multi-gigawatt manufacturing facility in Fremont, California, which represents a \$200 million investment, with the potential to create 400 new jobs. The company also operates a manufacturing facility in Newark, Delaware, where nearly 800 employees have assembled and shipped more than 20,000 Bloom Energy servers since 2013. Key to both facilities are nearly one thousand U.S. suppliers of components and services.

The fourth industrial revolution is underway, fueled by rapid advancements in AI and cloud computing. The rapid deployment of new data center capacity in the U.S. is a strategic priority, but there is a major bottleneck: power availability. Demand for power is only growing, while the electric grid is aging and new grid projects face permitting, transmission, distribution, and supply chain challenges.



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After 20 years of flat demand,¹ U.S. power needs are predicted to rise by 83 terawatt-hours (TWh) in 2025² – equivalent to powering an additional 7.7 million homes.³ The U.S. is expected to see the highest share of new data centers outside China. Since 2020, the U.S. colocation data center market has doubled, driven by digitization, cloud, and AI.⁴ To power this, data by 2030 could require 8-12% of the total U.S. demand compared to 3-4% today.⁵

Utilities face bottlenecks with transporting power via transmission and distribution infrastructure. As a result, grid interconnection takes longer, and capacity is increasingly expensive. If the U.S. continues building high-voltage transmission at its current rate, it will take at least 80 years to deliver what we need over the next decade.⁶ In fact, investment in new transmission by investor-owned utilities decreased from 2021 to 2023 when capital expenditures dropped from approx. \$9.2 billion to \$8.8 billion.⁷

To meet energy capacity needs the U.S. must promote technologies that expedite delivery of power (“Time to Power”). Routine grid upgrades needed to provide the required capacity to run new data centers on average take 48-60 months, whereas SOFCs can be operational in a fraction of that.

Onsite generators partner closely with utilities to reduce burdens on the collective power grid and transmission infrastructure. Historically, data centers have mostly used onsite power for backup. But data center and hyperscale developers have noted grid challenges and are increasingly turning to onsite power as a primary source. In January, President Trump highlighted his preference for onsite power to bolster AI infrastructure in a videoconference speech to the World Economic Forum.

To remain competitive in the AI Race, we must increase and incentivize the use one of the nearest term, most reliable, resilient onsite sources: domestically manufactured stationary fuel cells and microgrids. The AI Action Plan should include this recommendation.

SOFC Energy Servers provide mission critical reliability with no downtime for maintenance and are proven resilient through outages from hurricanes, earthquakes, winter storms, forest fires, other

¹ Grid Strategies, “Strategic Industries Surging: Driving U.S. Power Demand”

² International Energy Agency, “Electricity Mid-Year Update”

³ U.S. Energy Information Administration, “How much electricity does an American home use?”

⁴ Forbes, “Data Centers At The Heart Of The AI And Digital Economy Surge”

⁵ Lawrence Berkeley National Laboratory, “2024 United States Data Center Energy”; Goldman Sachs, “Generational growth: AI, data centers and the coming U.S. power demand surge”; McKinsey & Company, “How data centers and the energy sector can satiate AI’s hunger for power”

⁶ U.S. Department of Energy, “National Transmission Needs Study”; Americans for a Clean Energy Grid, “Transmission Projects Ready to Go: Plugging Into America’s Untapped Renewable Resources”; C Three Group via Robert Bryce, “Out of Transmission”

⁷ Edison Electric Institute, EEI Industry Capital Expenditures with Functional Detail (Oct. 2021, Sept. 2022, and Sept. 2023).



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extreme weather conditions, and natural disasters. In 2023, the average availability of Bloom Energy's fleet was 99.995%. Bloom has deployed more than 160 microgrids to date and our systems rode through 436 power outages carrying a total of 122,171 MWh of energy demand over 597 outage hours in 2022 alone.⁸ Bloom's technology provides substantial criteria pollutant, water, and greenhouse gas emission benefits, reducing bureaucratic delays associated with local air permitting.

Federal policies such as a 30% Investment Tax Credit for the purchase of fuel cells have helped keep this technology cost-competitive with cheaper but inferior fuel cells made in China. Unfortunately, efforts to ban the efficient use of natural gas from energy generation produced a regulation that made fuel cells ineligible for the ITC as of January 1, 2025. This is resulting in an unnecessary cost increase to deliver onsite energy for customers wishing to use U.S. natural gas as the feedstock for their domestically manufactured fuel cells.

Any final AI Action Plan should recommend that the National Economic Council work with Congress, and the Departments of Treasury and Energy to ensure U.S. manufactured fuel cells that efficiently utilize our country's vast supply of natural gas are part of a mix of onsite generation resources needed to secure American dominance in the AI Race.

Our country is poised to enter a new era of technological innovation and transformation. But as with any opportunity, obstacles must be overcome. In this case, our challenge is to provide the electricity required to grow a new and dynamic sector of the economy, and to do so with American-made energy solutions that will not generate exorbitant utility bills for families. The United States has a history of capturing such moments. Rising to the occasion is part of our DNA. With the right policies in place, we can not only lead the AI/data center revolution ahead of China, we can power it as well.

Thank you for the opportunity to comment on this Request for Information. We appreciate your consideration of our submission and welcome follow-up discussions or clarifications. Please contact me directly for any questions or if additional information is needed.

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Sincerely,

James Roth
Head of Global Policy
Bloom Energy

⁸ Bloom Energy, *How Bloom Reduces Emissions* (July 2024), <https://www.bloomenergy.com/wp-content/uploads/bloom-energy-how-bloom-reduces-emissions-technical-note.pdf>.



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